

EG1311: Design & Make

Course Information

Dr. Jason S. Ku



College of Design
and Engineering

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Logistics

- Lecture: LT3
- Lab: E4-02-06
- Website: Canvas



Account



Dashboard



Courses



Calendar





EG1311APC > Modules

Non-Academic

Home

People

Files

Assignments

Videos/Panopto

Collapse all

Export Course Content

▼ EG1311 E-Scholars APC

 Course Information

 Software

Staff

Lecturer: Dr. Jason Ku

Lab Staff: Mr. Vincent Bay Han Chuan

Lab A TAs: Ng Shi Qing, Eugenia
Yuan Wenxi

Lab B TAs: Darren Ng Zhen Yu
Koo Yao

Course Goals

- Core CDE Module: **Makerspace**
- Hands-on project-based engineering intro
- Design/build a robot to traverse course

Learning Outcomes

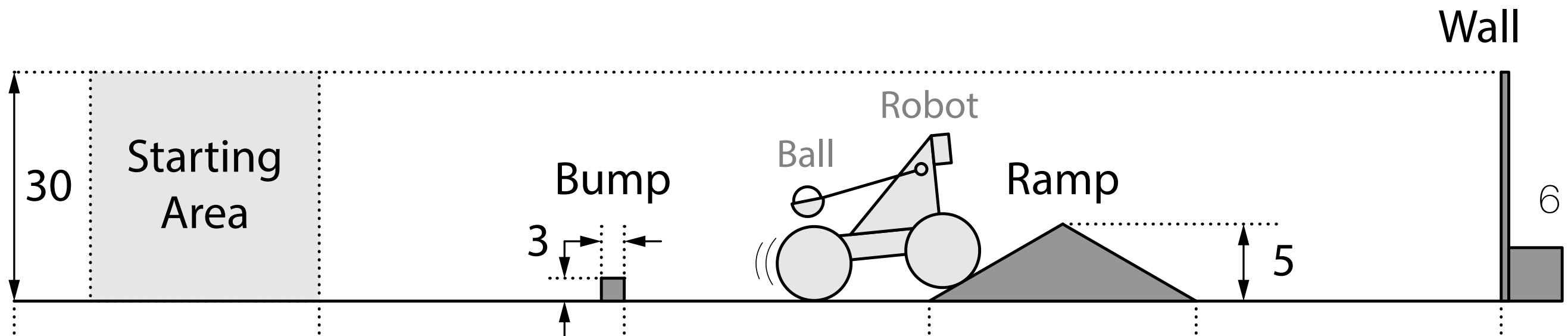
- Teamwork, communication, ideation
- Mechanical design, fabrication, CAD
- Electronics, Arduino, programming

Project



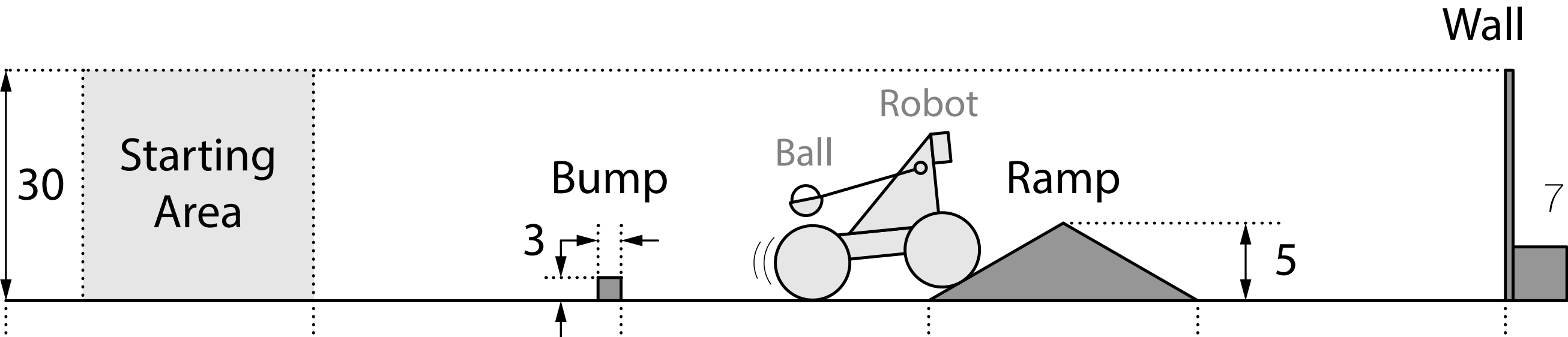
Project: Objective

- Robot start within 30 cm cube at start
- Must carry one 9V and six AA batteries
- Start in starting area with ping pong ball
- Traverse over obstacles to reach wall
- Stop within 5 cm of wall without touching it
- Deploy ball over wall without touching it



Project: Objective

Base Pts	Rubric Item
+1	All of the robot passes the bump
+1	All of the robot passes the ramp
+1	Part of the robot reaches within 5 cm of the wall
+1	No part of the robot touches the wall
+1	Robot is in contact with the ball after passing the ramp
+1	Ball leaves contact with robot after passing the ramp
+1	Ball goes over the wall
Weight Pts	Rubric Item
+1	Robot is not the heaviest among robots in the same section scoring all 7 base points
+1	Robot is the lightest among robots in the same section scoring all 7 base points
Extra Pt	Rubric Item
+1	After scoring all 7 base points, all of the robot returns behind the bump



Course Calendar

	M 20	T 21	W 22	R 23	F 24
9-11	Lec	A Lab	A Lab	A Lab	A Lab
12-2	Lec	Lec	Lec	Lec	Lec
3-5	Lec	B Lab	B Lab	B Lab	B Lab
	M 27	T 28	W 29		
9-12	A Lab	A Run	A Eval		
2-5	B Lab	B Run	B Eval		

Grading / Expectations

- Show up with a desire to learn & participate
- Most work done in class
- Pass/Fail, marks recorded internally
- Must pass all 4 components to pass class
- **Tue 28 July:** (1) Evaluated runs on course
- **Wed 29 July:** (2) In-class test
 - (3) 3-min Presentation
 - (4) 1 page Paper + Appendix

Software

- Arduino IDE (download + install)

<https://www.arduino.cc/en/software>

- TinkerCAD (online)

<https://www.tinkercad.com>

- Fusion (download + install)

<https://www.autodesk.com.sg/products/fusion-360>

Poll Everywhere

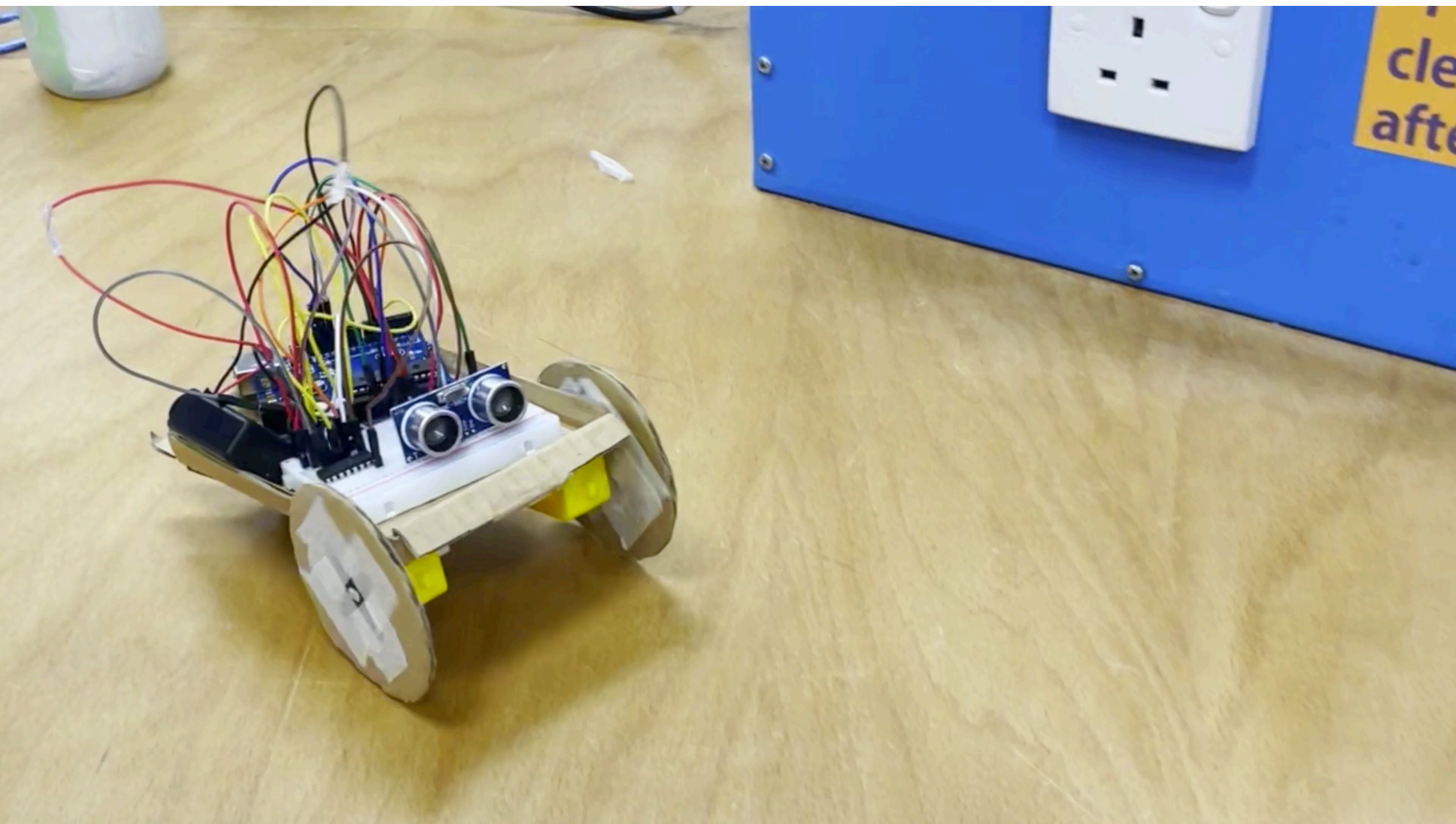
A

B

C

D

Template Project



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